Product Master — Product Type — Basic Design Document (基本設計書)

Document control (改版履歴)

Field	Value
Document ID	004_BD-TYPE
Category	UI + product reference data
System name	ProductionManagementAI
Subsystem name	Product master
Work item	WI-017
Based on brief.md revision	1
Created by	Claude (for ThanhTN)
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Version	Date	Author	Revision content
1	2026-10-25	Claude (for ThanhTN)	Initial creation: required product type 「種別」 on SCR-004 (WI-017 DEC-001–DEC-005)

This document adds one attribute to products and three places where SCR-004 製品マスタ (Product master) uses it. Everything it does not mention — search, state filter, paging, rows per page, retirement, unit locking, return after save, header and navigation — stays exactly as specified in 004_BD and as implemented by WI-013. Item numbers continue 004_BD's layout numbering (1–7), so new items start at 8.

System overview

Every product has exactly one type 「種別」 from a fixed list (DEC-001, DEC-004). Admins and Operators choose it when registering or editing a product, see it on the product list, and can narrow the list by it. Existing products get a type when the change is installed (DEC-002, DEC-005). Production orders, the dashboard and the order CSV export do not use the type (DEC-003).

Requirement ID	Description	Covered by section
REQ-090	Required type from a fixed list; existing rows backfilled	§3 Type values; Impact on existing data
REQ-091	Required 「種別」 select on register/edit	§1 item 10; §5; §6
REQ-092	「種別」 column (PC) and type on cards (SP)	§1 item 9
REQ-093	Type filter on the list, kept in the URL	§0-3; §1 item 8; §6

Overall configuration and architecture

No new component, endpoint or screen. The `products` table gains one required column; the existing product-master API carries the type in its create, update, detail and list payloads and accepts it as an optional list filter. The browser shows labels; the backend stores and validates codes. Exact column, constraint, migration and API changes are fixed in 004_DB-TYPE and 004_DD-TYPE.

Function list

Function ID	Function name	Description	Related requirement ID
FN-044	Classify a product	Store and validate a required type code on create and edit	REQ-090, REQ-091
FN-045	Filter the catalog by type	Return only products of the given type, combined with the existing search and state filters	REQ-093

Changed, not replaced: FN-027 (search and page), FN-028 (create) and FN-029 (edit) from 004_BD carry the type.

Actors and business flow

Actors: Admin, Operator (unchanged; no new permission).

Register a product with a type (extends 004_BD create flow)

1. User presses 「製品を登録」 → the form shows 「種別」 (Type) with 「選択してください」 (Please select) and the 「必須」 marker.
2. User selects e.g. 「部品」 and fills the other fields → 「保存」.
3. System validates the type with the other fields and saves; the list shows the new row with 「部品」.

Change a product's type: as above in edit mode. The type stays editable for retired products and for products referenced by orders, because orders do not depend on it (DEC-003).

Narrow the list by type: user picks 「種別」 = 「原材料」 in the filter panel and presses 「検索」 → the list shows only raw materials, page 1. 「クリア」 resets it to 「すべて」.

Screen list and screen transition

No new screen, route or transition; see 004_BD. The list URL gains the `type` parameter (§0-3).

Screen design detail

SCR-004 Product master — type additions

0-1. Basic information

Unchanged (004_BD 0-1).

0-3. URL parameters

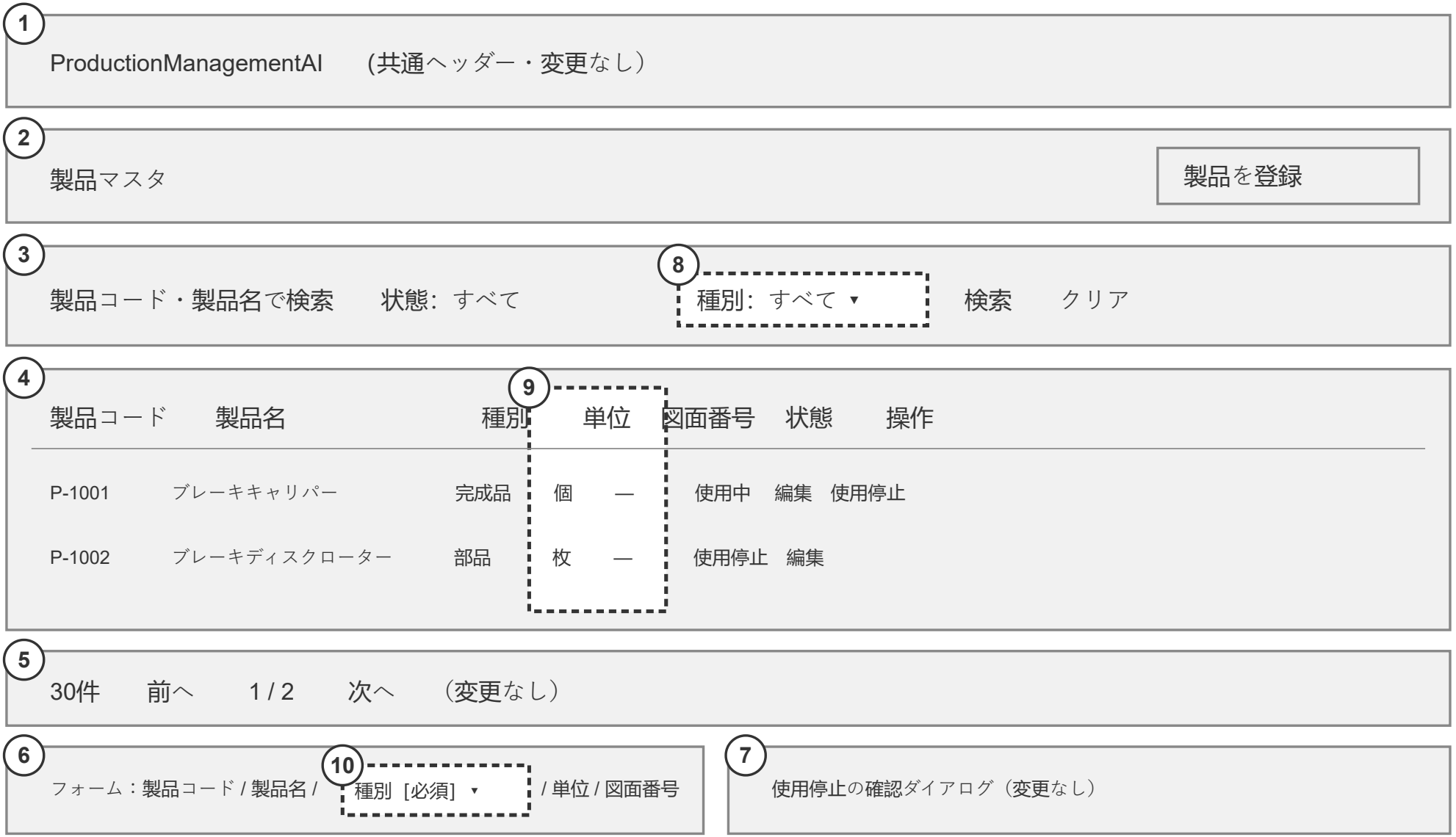
No	Parameter	Values	Default / invalid
1	`type` (new)	One type code from §3	Absent = all types. An unknown value is dropped from the URL and treated as all, like other invalid list parameters (WI-013)

Search, Clear, paging, rows per page and return-after-save keep `type` exactly as they keep the existing parameters; Clear removes it.

1. Layout and mockup

Layout sketch only; the rendered mockup belongs in 004_DD-TYPE. Dashed boxes are new.

PC / desktop



SP / mobile

1

ProductionManagementAI　メニュー

2

製品マスタ　［製品を登録］

3

製品コード・製品名で検索

状態: すべて

8

種別: すべて ▾

［検索］　［クリア］

4

P-1001　ブレーキキャリパー

9

種別: 完成品　単位: 個　使用中

［編集］　［使用停止］

5

30件　前へ　1 / 2　次へ

6

フォーム（1列）: 製品コード / 製品名

10

種別 [必須] ▾

単位 / 図面番号　保存 / キャンセル

Item No.	Region / element	Notes
1–2, 5, 7	Header, heading, paging, retire dialog	Unchanged
3	Search/filter	Unchanged, plus item 8
8	「種別」 filter (new)	Select: 「すべて」 (All) + the four labels in §3 order; placed after 「状態」 (State). Applied only on 「検索」, like the state filter
4	Catalog	Unchanged, plus item 9
9	「種別」 column / card line (new)	PC: column after 「製品名」 (Name), label text. SP: 「種別：完成品」 on the card line before 「単位：」. Not sortable (the list has no column sorting)
6	Product form	Unchanged, plus item 10
10	「種別」 select (new)	Required; after 「製品名」, before 「単位」. Create: placeholder 「選択してください」, no preselection. Edit: current value

2. Content blocks and visible states

No new state. Validation failure, conflict and success states (004_BD §2) include the type field. A no-match result caused by the type filter uses the existing no-match message.

3. Screen item definition — type values

Order	Code	Japanese UI text (English gloss)
1	`FINISHED`	「完成品」 (Finished good)
2	`SEMI_FINISHED`	「半製品」 (Semi-finished good)
3	`PART`	「部品」 (Part)
4	`RAW_MATERIAL`	「原材料」 (Raw material)

Codes are stored and exchanged; labels are display only. Changing the list later is a schema change (DEC-001).

4. Item value mapping

Value	Display
Type code	Label from §3
Filter not set	「すべて」 (All)

Value	Display
Form, no type chosen (create)	「選択してください」 (Please select)

5. Validation checks

Check	User result	Requirement
Type not chosen	Field error 「種別を選択してください。」 (Please select a type.), in the error summary; save blocked	REQ-091
Unknown type code (API)	400 validation error on the type field; backend rejects independently of the UI	REQ-090
Unknown `type` list filter (API)	400 validation error on `type`; the UI never sends one (§0-3)	REQ-093

Message IDs are assigned in 004_DD-TYPE.

6. Item events

Event	Result
Change 「種別」 filter	No request until 「検索」 (same as 「状態」)
「検索」 / 「クリア」	Update URL `type` (set / remove) and fetch page 1
Change 「種別」 in form	Marks the form dirty (existing unsaved-change guard)
Save	Type sent with the other fields; field error kept on failure

Impact on existing screens and data

- **Existing products:** the migration sets every existing product to `FINISHED` (「完成品」) (DEC-005). The demo catalog instead gets a realistic type per product, listed in 004_DB-TYPE.
- **SCR-001, SCR-002, SCR-003, order CSV:** no change (DEC-003). `GET /api/products` (order screens' product choices) does not gain the type.
- **Tests and fixtures** that create products must send a type.

Non-functional design

- **Accessibility:** the filter and form selects have visible labels; the required marker is text, not colour only; the type error is linked to the field (as other fields, WCAG 2.2 AA).

- **Security:** unchanged roles; server-side allow-list validation of the code on writes and on the filter.
- **Performance:** four values on a small catalog; whether the filter needs an index is decided in 004_DB-TYPE.
- **Observability:** no new telemetry; type values are not sensitive.